Residual Phase Clocks and Branch Certificates for the Planar XY Essential-Singularity Flow

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

The assigned October 5 manuscript claims a finite positive essential-singularity amplitude for the ordinary square-lattice cosine XY model. We read its full cached text and isolate the deterministic passage-time problem from the analytic entry, temperature transversality, height-profile and mass-transfer arguments. For the hyperbolic branch of the balanced quadratic flow we prove an integrated-residual phase certificate, with explicit invariant margins. For the actual quadratic Euler map we obtain a corrected discrete clock: the phase increment divided by the incoming invariant radius, plus half the logarithmic amplitude increment, differs from one by at most seven times the squared amplitude. We extend this estimate to perturbed steps with a branch-aware, non-exponential error ledger and give finite rational panel certificates. An explicit two-constants radius and a Cauchy secant budget clarify the source's shrinking analytic domains. These are local deterministic refinements, not certification of the full renormalisation map, its transverse temperature coefficient, or the probabilistic comparison with spin mass. Contribution assessment: C3/5, the corrected discrete clock and its explicit branch-aware residual extension are notable source-specific local refinements, not an independent proof of the XY essential-singularity claim.

Contents

1 Model, normalization and status **2**

2 What the source actually has to prove **3**
2.1 Balanced coordinates and the transverse parameter 3
2.2 Exit, profile, and the two mass inequalities 4

3 An integrated residual clock **4**

4 A corrected clock for the actual quadratic step **6**
4.1 Invariant bookkeeping before phase bookkeeping 7

5 Finite panels and reproducible branch certificates **8**
5.1 An activity-memory budget 9
5.2 Scaling and a conditional uniform application 9

6 Shrinking analytic disks: an explicit budget **10**

7 From original Ram proofs to this report **11**

8 Exit and mass interfaces that remain open inputs **12**

9 Advantages, costs and next mathematical steps **12**

10 Evidence and contribution boundary **13**

1 Model, normalization and status

For a finite free square box $\Lambda \subset \mathbb{Z}^2$, the source uses

$$\frac{1}{Z_{\Lambda,b}} \exp\left(b \sum_{\{u,v\}\in E(\Lambda)} \cos(\theta_u - \theta_v) \right) \prod_{u\in\Lambda} \frac{\mathrm{d}\theta_u}{2\pi}, \qquad b > 0, \quad \theta_u \in \mathbb{R}/(2\pi\mathbb{Z}).$$

Each unoriented nearest-neighbour edge occurs once. The thermodynamic limit is taken before the separation limit:

$$C_b(r) = \lim_{R\to\infty} \mathbb{E}_{\Lambda_R,b} \cos(\theta_0 - \theta_{re_1}), \qquad m(b) = \lim_{r\to\infty} -r^{-1} \log C_b(r), \quad \xi(b) = m(b)^{-1}.$$

With $b_c = \inf\{b : m(b) = 0\}$, the assigned source [1] claims

$$\lim_{b\uparrow b_c} \sqrt{b_c - b} \log \xi(b) = A_{\mathrm{XY}} \in (0, \infty). \tag{1}$$

Here and below an unadorned logarithm is natural. The local theorems of this report do not establish (1).

The source height has spacing 2π: its edge integer k has weight $e^{-b} I_k(b)$, equivalently the difference of two independent Poisson variables of mean $b/2$. Its Laplacian uses unit-edge energy $\sum_{\{u,v\}} (h_u - h_v)^2$, not one quarter of that energy. The reference critical Gaussian coefficient is $a = 8\pi$ in these height units. Neither changing the height spacing nor changing the RG blocking factor permits an untracked change of the amplitude in (1).

Current primary context. Lammers' Theorem 1 [3], with its definitions of the two masses, states $\xi_{\mathrm{XY}} = 2\xi_{\mathrm{Height}}$ at each fixed inverse temperature. This is a mass relation, not a formula for its rate of divergence at criticality. Durand–Lammers' May 2026 preprint, Theorem 1.4 [4], states, for its integer-height free energy and $0 < s < 1/1000$,

$$s\, m_{\mathrm{XY}}(b) \le f_b(s) - f_b(0) \le s\, m_{\mathrm{XY}}(b) + cs^2/2$$

with a universal c. This supplies a different quantitative mass interface, but not the amplitude asserted in (1). We inspected these primary statements and conventions, not their complete external proofs. Their height variable is integer-valued; the assigned source multiplies that height by 2π. A fixed nonzero multiplicative covariance factor does not change an exponential decay rate, whereas a slope or energy coefficient must be rescaled explicitly. Literature inspection here is current to October 9, 2026 and is not an exhaustive novelty determination.

2 What the source actually has to prove

The source is substantially more than an integration of the familiar quadratic flow. Table 1 records its complete scientific architecture, including portions outside our new local deductions. All listed source files were read literally in bounded displays.

Table 1: Source modules and retained obligations.

Module	Mechanism and boundary
`finite.tex`, 291 lines	Finite-volume Gaussian/lattice comparisons, Poisson unit insertion, complex likelihood control on $\lvert b-b_c\rvert\,\lvert E\rvert \ll 1$, and a strictly positive temperature response. Positivity and boundary conventions cannot be replaced by an arbitrary covariance approximation.
`entry.tex`, 862 lines	Scale-s analytic initialization on a disk of radius $c_{R_0}s^{-2}$; local normalizers, winding links, field regulators, common Gaussian completion and two-constants interpolation. The fixed analytic RG map and critical asymptotics import companion results.
`trajectory.tex`, 769 lines	Derivative slaving, canonical temperature vector with positive coefficient P, comparison of different entry scales through physical readouts, late-entry secants, scaled quadratic ODE and outgoing-cone exit.
`profile.tex`, 1369 lines	Moving-temperature height profiles; Gaussian identification, phase vanishing, finite pins, band tilts and ordered limits; exclusion of intermediate positive profile values. This is not a finite-dimensional consequence of the flow clock.
`mass.tex`, 610 lines	Spin/loop estimates for a lower correlation-length bound; twisted annuli and a Lieb–Rivasseau finite-size criterion for the upper bound. Actual spin mass is identified only after both directions.

2.1 Balanced coordinates and the transverse parameter

The source fixes a blocking factor $L > 1$, a small tube and regulator constants before the near-critical limit. Its balanced state is (W_j, Q_j) with $W_j = (x_j, y_j) \in \mathbb{R}^2$ and Q_j in a scale-dependent activity Banach space. Suppressing fixed norm constants, its estimates have the form

$$W_{j+1} = W_j + F(W_j) + e_j, \qquad F(x,y) = (-y^2, -xy), \tag{2}$$

$$|e_j| \le C\{j^{-2}|W_j|^2 + |W_j|^3 + (|W_j| + \|Q_j\|)\,\|Q_j\|\}, \tag{3}$$

$$\|Q_{j+1}\| \le \vartheta\,\|Q_j\| + C|W_j|^2, \quad 0 < \vartheta < 1. \tag{4}$$

The source also has torus-copy estimates in their own spaces. Different entry-scale remainder spaces are not silently subtracted. The physical readout comparison is what permits comparison of their two-dimensional balanced coordinates.

At criticality $W_j = (1,1)/j + O(\log j/j^2)$, and the source obtains a canonical derivative

$$\frac{\partial_b W_j(b_c)}{j} \longrightarrow P(1,-1/2), \qquad P > 0. \tag{5}$$

The matrix governing the linearized leading flow is $\left(\begin{smallmatrix} 0 & -2 \\ -1 & -1 \end{smallmatrix}\right)$, with eigenvalues $1, -2$. Slaving alone does not prove $P > 0$; the source uses strict finite-volume response and an unstable-cone argument. Neither the residual theorems below nor analytic uniqueness supplies that strict transverse input.

Put $\varepsilon = b_c - b$, $\delta = \sqrt{\varepsilon}$ and $w = \delta j$. The source's late entry has $s \asymp \varepsilon^{-1/3}$ and $J \asymp \log s$, so

$$\varepsilon s^2 \to 0, \qquad \delta^{-1} = o(s^2), \qquad J = o(\delta^{-1}). \tag{6}$$

On compact subintervals of the positive-invariant branch its limiting coordinates solve

$$X' = -Y^2, \qquad Y' = -XY, \qquad Y^2 - X^2 = 3P =: \kappa^2 > 0. \tag{7}$$

The initial asymptotic selects

$$X(w) = \kappa \cot(\kappa w), \quad Y(w) = \kappa \csc(\kappa w), \quad 0 < w < T := \pi/\kappa. \tag{8}$$

For example, $X = 1/w - \kappa^2 w/3 + O(w^3)$ and $Y = 1/w + \kappa^2 w/6 + O(w^3)$ match the vector in (5). The endpoint $w = T$ is a pole, not a point where a compact-window ODE error estimate can simply be evaluated.

2.2 Exit, profile, and the two mass inequalities

The source proves exit at $x = -d + O(d^2)$, $y = O(d)$ and $Q = O(d^2)$, with fixed small $d > 0$ and $\delta j_e \to T$. Its outgoing cone controls $-x/y \in [1/2, 2]$ and $y + cy^2 \le y_{\text{next}} \le y + Cy^2$; it first fixes a large scaled amplitude M, then lets $\delta \downarrow 0$, and only afterwards lets $M \to \infty$. This ordering is retained.

Separately, its affine height profile is 8π before T and zero after T, with no assertion needed at T itself. Positive-profile pinned Gaussian covariance, together with a spin-loop exponential bound, gives the lower mass-scale comparison. Zero-profile annuli, a fixed tilt followed by mesh refinement, and a finite-size spin inequality give the upper comparison. The many limits in the pinned argument include cable subdivision, primary mesh, band mesh and band strength in specified order. They are not a numerical continuation certificate. Only after these inputs does the source conclude

$$\delta \log_L \xi(b) \longrightarrow T, \qquad A_{\mathrm{XY}} = \frac{\pi \log L}{\sqrt{3P}}. \tag{9}$$

We preserve these mechanisms but do not independently certify their global uniform estimates or the companion critical-exponent and logarithmic-correction papers [2].

3 An integrated residual clock

Here all assertions are deterministic and self-contained. Write

$$\mathcal{H} = \{(x, y) \in \mathbb{R}^2 : y > |x|\}.$$

For an absolutely continuous path $z(t) = (x(t), y(t))$ on a compact interval $I = [t_0, t_1]$, define the actual defects, almost everywhere, by

$$r_x = x' + y^2, \qquad r_y = y' + xy.$$

There is no assumption that a sampled defect bounds the true defect. Put $K = y^2 - x^2$, $k = \sqrt{K}$ when $K > 0$, and $\theta = \operatorname{atan2}(k, x) \in (0, \pi)$ when $z \in \mathcal{H}$. The angle is defined by $\cos\theta = x/y$ and $\sin\theta = k/y$; no complex logarithm branch is being continued through zero.

Theorem 3.1 (Positive-branch residual clock)**.** *Suppose $z(t_0) \in \mathcal{H}$ and a certified number B satisfies*

$$2\int_I (|yr_y| + |xr_x|)\,\mathrm{d}t \le B < K(t_0).$$

Then the whole path belongs to $\mathcal{H}$. Define

$$k_- = \sqrt{K(t_0) - B}, \quad k_+ = \sqrt{K(t_0) + B}, \quad E = \frac{1}{k_-}\int_I (|r_x| + |r_y|)\,\mathrm{d}t.$$

With $\Delta\theta = \theta(t_1) - \theta(t_0)$ and $\tau = t_1 - t_0$,

$$\max\left\{0, \frac{\Delta\theta - E}{k_+}\right\} \le \tau \le \frac{\Delta\theta + E}{k_-}. \tag{10}$$

More generally the same bound holds whenever a whole-path certificate gives $0 < k_- \le k(t) \le k_+$ and $y(t) > 0$, without using the displayed B.

Proof. The polynomial chain rule, valid also before branch positivity is known, gives

$$K' = 2yr_y - 2xr_x. \tag{11}$$

Thus $K(t) \in [K(t_0) - B, K(t_0) + B]$. Since $K > 0$, the continuous y cannot cross zero, and its initially positive sign persists. On this compact path the angle is a smooth function. Direct differentiation gives

$$\partial_x\theta = -\frac{1}{k}, \qquad \partial_y\theta = \frac{x}{yk}, \qquad \theta' = k + \frac{(x/y)r_y - r_x}{k}. \tag{12}$$

The last residual term has integral of absolute value at most E, since $|x/y| < 1$. Hence $k_-\tau - E \le \Delta\theta \le k_+\tau + E$. Rearrangement and $\tau \ge 0$ prove the result. □

Corollary 3.2 (A duration bound without an integrated unknown)**.** *Under a whole-path branch bound, if $(|r_x| + |r_y|)/k_- \le \eta < k_-$ almost everywhere, then θ is strictly increasing and*

$$\frac{\Delta\theta}{k_+ + \eta} \le \tau \le \frac{\Delta\theta}{k_- - \eta}.$$

Proof. Equation (12) places θ' between $k_- - \eta > 0$ and $k_+ + \eta$. Integrate. □

The theorem uses an integrated, geometrically weighted error, rather than $e^{L\tau}$ times a sup-norm defect. Its cost is the lower bound on K. It is not uniform as $K \downarrow 0$. On a fixed scaled near-critical window, K is close to $3P > 0$ if that assertion has separately been certified; in unscaled coordinates K is of order δ^2. Using the same absolute tolerance in both coordinate systems is incorrect.

Example 3.3 (A signed bias that a scalar norm conceals)**.** Suppose the path solves $x' = -(1+\lambda(t))y^2$, $y' = -(1 + \lambda(t))xy$ with $1 + \lambda(t) > 0$ integrable. Then $K' = 0$ exactly and $\theta' = k(1 + \lambda(t))$. Consequently $\Delta\theta/k = \tau + \int_I \lambda(t)\,\mathrm{d}t$. The separate absolute-defect bound is valid, but the signed formula retains cancellations and the invariant has no loss at all. This is a conditional time reparametrization example, not a model of all RG errors in (3).

4 A corrected clock for the actual quadratic step

The continuous invariant is not exactly preserved by the discrete quadratic part of (2). This matters before adding any activity remainder. Let

$$\Phi(x,y) = (x - y^2, y - xy), \qquad 0 < y \le \tfrac{1}{4}, \quad |x| < y.$$

Lemma 4.1 (Invariant and angle of one Euler step)**.** *Writing $k = \sqrt{y^2 - x^2}$ and $s = \sqrt{1-y^2}$, the new invariant radius is $k^+ = ks$, and $\Phi(x,y) \in \mathcal{H}$. If*

$$D = 1 - x - \frac{k^2}{1+s}, \qquad H = 1 - \frac{x}{1+s}, \qquad h = H/D, \tag{13}$$

then

$$\theta^+ - \theta = \arctan(kh) \in (0, \pi/2), \qquad D \ge 11/16, \quad 3/5 \le h < 2. \tag{14}$$

Proof. Expansion gives the exact identity

$$(y - xy)^2 - (x - y^2)^2 = (y^2 - x^2)(1 - y^2). \tag{15}$$

Also $y^+ = y(1-x) > 0$. Multiplication of the complex numbers encoding the two angles yields

$$\frac{(x - y^2 + iks)(x - ik)}{y^2} = D + ikH.$$

This uses $1 - s = y^2/(1+s)$ and is an identity, not a small-angle approximation. Since $D \ge 1 - |x| - k^2 \ge 11/16$ and $H \in [3/4, 5/4]$, the phase difference is positive and less than $\pi/2$. Furthermore $D \le 5/4$, giving $h \ge 3/5$, while $h \le (5/4)/(11/16) = 20/11 < 2$. □

Theorem 4.2 (Quadratic-error corrected discrete clock)**.** *For every state in Lemma 4.1,*

$$\left| \frac{\theta^+ - \theta}{k} + \frac{1}{2} \log \frac{y^+}{y} - 1 \right| \le 7y^2. \tag{16}$$

Consequently, along any N successive exact quadratic steps which stay in $0 < y_j \le 1/4$, $|x_j| < y_j$,

$$\left| \sum_{j=0}^{N-1} \frac{\theta_{j+1} - \theta_j}{k_j} + \frac{1}{2} \log \frac{y_N}{y_0} - N \right| \le 7 \sum_{j=0}^{N-1} y_j^2. \tag{17}$$

Proof. The exact numerator of $h - (1 + x/2)$ is

$$H - D(1 + x/2) = -\frac{x(1-s)}{2(1+s)} + \frac{x^2}{2} + \frac{k^2(1 + x/2)}{1+s}. \tag{18}$$

Its absolute value is at most $y^3/2 + y^2/2 + (9/8)y^2 \le (7/4)y^2$. Division by $D \ge 11/16$ gives $|h - 1 - x/2| \le (28/11)y^2 < 3y^2$. For $u \ge 0$, $0 \le u - \arctan u = \int_0^u t^2/(1+t^2)\,\mathrm{d}t \le u^3/3$. Therefore

$$\left| \frac{\arctan(kh)}{k} - 1 - \frac{x}{2} \right| \le 3y^2 + \frac{k^2 h^3}{3} < \frac{17}{3} y^2.$$

The power series, or integration of $t/(1-t)$, gives $|\log(1-x) + x| \le x^2/[2(1-|x|)]$. Thus the extra half-logarithm cancels $x/2$ with error at most $y^2/3$. The total is at most $6y^2$, and hence satisfies the slightly relaxed constant 7 in (16). Summation telescopes the logarithms. □

The slack in the stated constant leaves room for convenient rational majorants; no optimal constant is claimed. Without the logarithmic correction, the normalized phase increment is $1 + x/2 + O(y^2)$. For a certified window with $y_j \le C/j$, the corrected leading-map budget is at most $7C^2/(J-1)$ over $j \ge J > 1$, rather than a bound proportional to $\sum |x_j|$. For a window with $y_j = O(\delta)$ and $N = O(\delta^{-1})$, it is $O(\delta)$. These are conditional deterministic improvements of the error ledger; they do not erase the actual defects in (3).

4.1 Invariant bookkeeping before phase bookkeeping

For a perturbed step $z_{j+1} = \Phi(z_j) + e_j$, write $(\bar{x}_j, \bar{y}_j) = \Phi(z_j)$ and $e_j = (e_{x,j}, e_{y,j})$. Then

$$\begin{aligned} K_{j+1} &= (1 - y_j^2)K_j + d_j, \\ d_j &= 2\bar{y}_j e_{y,j} - 2\bar{x}_j e_{x,j} + e_{y,j}^2 - e_{x,j}^2. \end{aligned} \tag{19}$$

If $a_j = 1 - y_j^2 > 0$ and $P_n = \prod_{j<n} a_j$, induction gives

$$K_n = P_n \left(K_0 + \sum_{j<n} \frac{d_j}{P_{j+1}} \right). \tag{20}$$

Certified signed intervals for d_j propagate through this formula; discarding their signs gives the sufficient condition $K_0 > \sum_{j<n} |d_j|/P_{j+1}$ for positivity up to time n. One must also retain $y_n > 0$. In the step theorem below it is checked directly. A positive invariant by itself allows either the upper or the lower cone.

Theorem 4.3 (Perturbed clock with an explicit branch cost)**.** *Let $z_0, \dots, z_N$ be actual states, with $0 < y_j \le 1/4$, $|x_j| < y_j$ for $j < N$, and $z_N \in \mathcal{H}$. Set $e_j = z_{j+1} - \Phi(z_j)$. On the segment from $\Phi(z_j)$ to z_{j+1} define the certified bounds*

$$\begin{aligned} a_j^- &= \min\{\bar{y}_j - \bar{x}_j, y_{j+1} - x_{j+1}\}, & a_j^+ &= \min\{\bar{y}_j + \bar{x}_j, y_{j+1} + x_{j+1}\}, \\ \underline{k}_j &= \sqrt{a_j^- a_j^+}, & \underline{y}_j &= \min\{\bar{y}_j, y_{j+1}\}. \end{aligned}$$

These are strictly positive. With $k_j = \sqrt{K_j}$ put

$$B_N = \sum_{j<N} \left(7y_j^2 + \frac{|e_{x,j}| + |e_{y,j}|}{k_j \underline{k}_j} + \frac{|e_{y,j}|}{2\underline{y}_j} \right). \tag{21}$$

Then

$$\left| \sum_{j<N} \frac{\theta_{j+1} - \theta_j}{k_j} + \frac{1}{2} \log \frac{y_N}{y_0} - N \right| \le B_N. \tag{22}$$

If additionally

$$\frac{|e_{x,j}| + |e_{y,j}|}{\underline{k}_j} \le \frac{2k_j}{5} \quad (j < N), \tag{23}$$

and $0 < k_- \le k_j \le k_+$, then $\theta_{j+1} \ge \theta_j$ and

$$\max\left\{ 0, \frac{\theta_N - \theta_0}{k_+} + \frac{1}{2} \log \frac{y_N}{y_0} - B_N \right\} \le N \le \frac{\theta_N - \theta_0}{k_-} + \frac{1}{2} \log \frac{y_N}{y_0} + B_N. \tag{24}$$

Proof. The cone $\mathcal{H}$ is convex. Each factor $y - x$ and $y + x$ is affine on the comparison segment and bounded below by a_j^- and a_j^+. Thus $k \geq \underline{k}_j$ and $y \geq \underline{y}_j$ there. Equation (12) and the mean-value integral bound the phase change caused by e_j by $(|e_{x,j}| + |e_{y,j}|)/\underline{k}_j$; the logarithmic change is at most $|e_{y,j}|/\underline{y}_j$. Apply Theorem 4.2, divide the phase error by the incoming k_j, and sum.

For monotonicity, $\arctan u \geq u/(1 + u^2)$ for $u \geq 0$, by integration or differentiation. Lemma 4.1 gives $h \geq 3/5$ and $kh \leq 1/2$, so the unperturbed increment is at least $12k_j/25$. Subtracting at most $2k_j/5$ still leaves $2k_j/25 > 0$. Every increment is therefore nonnegative. Only now may the weighted phase sum be bounded between $(\theta_N - \theta_0)/k_+$ and $(\theta_N - \theta_0)/k_-$. This and (22) yield (24). □

Why the conditions are not cosmetic. Without (23), (22) still holds, but bounding a signed weighted phase sum by just its endpoint difference is generally invalid. Without the segment branch bounds, a small Euclidean perturbation near $K = 0$ can cross the separatrix and no real phase clock remains. The theorem does not presume that a solver's floating-point orbit is the actual RG orbit.

5 Finite panels and reproducible branch certificates

The previous theorems become implementable when the path, or enclosures of its true values, are specified on complete panels. Values at finitely many isolated parameter points are insufficient.

Proposition 5.1 (Exact affine-panel residuals)**.** *Let $h > 0$ and let $z(t)$ be affine between (x_0, y_0) and (x_1, y_1) on $0 \leq t \leq h$. Write $u = t/h$, $v_x = (x_1 - x_0)/h$ and $v_y = (y_1 - y_0)/h$. In the degree-two Bernstein basis $(1-u)^2, 2u(1-u), u^2$, the coefficients of its true defects are*

$$r_x : (v_x + y_0^2,\ v_x + y_0 y_1,\ v_x + y_1^2), \tag{25}$$

$$r_y : (v_y + x_0 y_0,\ v_y + (x_0 y_1 + x_1 y_0)/2,\ v_y + x_1 y_1). \tag{26}$$

If both endpoints lie in $\mathcal{H}$, the whole panel does. With $a^\pm = \min\{y_0 \pm x_0, y_1 \pm x_1\}$ it satisfies $K(t) \geq a^- a^+ > 0$. The coefficient triple of K is

$$(y_0^2 - x_0^2,\ y_0 y_1 - x_0 x_1,\ y_1^2 - x_1^2). \tag{27}$$

The minimum of this triple is a further valid, possibly weaker, lower bound. The maximum absolute coefficient of either defect bounds its absolute value everywhere, and its integral is at most h times that bound.

Proof. Expand the products of the affine endpoint interpolants. Bernstein basis functions are nonnegative and sum to one. The factors $y - x$ and $y + x$ remain above their positive endpoint minima, proving the branch statement without needing positivity of every coefficient in (27). Integration of a uniform bound proves the last claim. □

The Bernstein hull property used here is classical. The useful local content is the explicit defect and branch module for this particular flow, including both the quadratic invariant and the normalized phase. For exact rational endpoints and h, every coefficient and the squared branch lower bound is rational. Rational lower bounds on the square roots can be certified by squaring nonnegative numbers. Angles and logarithms at the final stage require proved transcendental enclosures if numerical endpoints are wanted; their exact symbolic appearance in the theorems does not require numerical evaluation.

Example 5.2 (An entire panel, not a sampled residual)**.** Take $z(t) = (-t, 1)$ on $0 \le t \le h < 1$. Then $r_x = 0$, $r_y = -t$, $K = 1 - t^2$, $B = h^2$ and $\int(|r_x| + |r_y|) = h^2/2$. Theorem 3.1 applies with $k_- = \sqrt{1-h^2}$, $k_+ = 1$ (using the sharper signed invariant bound) and $\Delta\theta = \arcsin h$. For $h = 1/10$ all pre-angle budgets are exact rational quantities; the time is, of course, exactly $1/10$. This fixture checks a nonzero-defect path and is not a sampled XY trajectory.

Example 5.3 (One perturbed discrete step)**.** Take

$$z_0 = (1/16, 1/8), \quad \Phi(z_0) = (3/64, 15/128), \quad e = (0, 1/4096).$$

Then $z_1 = (192/4096, 481/4096)$ lies in $\mathcal{H}$, $k_0^2 = 3/256$, and the segment lower square is $189/16384$. Both k_0 and $\underline{k}_0$ exceed $1/10$ by exact squaring. The phase perturbation is at most $10/4096 < 1/25 \le 2k_0/5$. Thus the monotonicity gate is certified. Formula (21) admits the purely rational upper budget

$$B_1 \le \frac{7}{64} + \frac{25}{1024} + \frac{1}{960}.$$

This deliberately simple bound is not asserted optimal. It shows exactly which positivity and normalization costs a rational certificate must pay.

5.1 An activity-memory budget

To apply a finite clock to (2), the actual infinite-dimensional remainder must still be enclosed. The contraction estimate (4) implies by induction

$$\|Q_j\| \le \vartheta^{j-J} \|Q_J\| + C \sum_{i=J}^{j-1} \vartheta^{j-1-i} |W_i|^2. \tag{28}$$

If the older states satisfy $|W_i| \le M$, omitting all but the last H summands loses at most $CM^2\vartheta^H/(1-\vartheta)$. The initial term must still be kept. Inserting a certified upper bound from (28) into (3) produces componentwise step-defect bounds after the chosen norm comparison is accounted for. This is the ordinary geometric convolution bound, credited as such. It is not legitimate to obtain (4) by inspecting a finite matrix representation of only a few activities.

5.2 Scaling and a conditional uniform application

Suppose, on a fixed compact w-interval, a family of actual RG states has certified bounds

$$|W_j| \le C_0\delta, \quad \|Q_j\| \le C_1\delta^2, \quad j \asymp \delta^{-1}, \quad k_j \ge c_0\delta,$$

and comparison segments also have $\underline{k}_j \ge c_1\delta$ and $\underline{y}_j \ge c_2\delta$, with positive constants independent of δ. Formula (3) gives $|e_j| = O(\delta^3)$. There are $O(\delta^{-1})$ steps. Consequently

$$\sum y_j^2 = O(\delta), \quad \sum \frac{|e_j|_1}{k_j \underline{k}_j} = O(1), \quad \sum \frac{|e_{y,j}|}{\underline{y}_j} = O(\delta).$$

The perturbed clock has an $O(1)$ error in the number of steps, hence an $O(\delta)$ error after multiplication by δ. This statement controls its explicit clock remainder, not every other quantity in the endpoint bracket. In particular, errors in $k_\pm$, endpoint phases, entry data and the mass interface must be added separately. For small enough δ the monotonicity condition follows from the same uniform constants, since its two sides have orders δ^2 and δ.

Equivalently, the affine interpolant of W_j/δ at times $w = \delta j$ has defects of order δ on each whole panel. This follows from $e_j/\delta^2 = O(\delta)$ and the quadratic variation of F within a panel; Proposition 5.1 gives exact coefficients. The continuous certificate then has an integrated $O(\delta)$ budget on a compact interval with a positive scaled invariant margin. Neither formulation covers the pole of (8), where boundedness and uniform branch constants require a separate outgoing argument.

6 Shrinking analytic disks: an explicit budget

The original Ram continuation discussion [6] distinguishes analytic uniqueness from existence and stable inversion. The source respects this distinction: its small complex activity norm is obtained by a majorant, a real gain and two-constants interpolation, not by an identity theorem alone. Here is an explicit version of that interpolation for its size-indexed pattern bounds.

Proposition 6.1 (A computable inner disk)**.** *Let $\mathcal{B}$ be a fixed complex Banach space, and let $f_\ell : D_R \to \mathcal{B}$ be holomorphic for every positive integer ℓ. Suppose*

$$\|f_\ell(z)\| \le e^{b\ell} \quad (|z| < R), \qquad \|f_\ell(t)\| \le e^{-a\ell} \quad (t \in (-R, R)), \quad a > c > 0,\ b \ge 0.$$

For any $0 < \vartheta < 1$ with

$$\frac{4}{\pi} \arctan \vartheta \le \frac{a-c}{a+b}, \tag{29}$$

one has $\|f_\ell(z)\| \le e^{-c\ell}$ for $|z| \le \vartheta R$. In particular the right side gives an explicit permissible ϑ by taking its tangent after multiplication by $\pi/4$.

Proof. In the upper half unit disk the harmonic measure of the semicircular arc is

$$\omega(z) = \frac{2}{\pi} \arg \frac{1+z}{1-z}.$$

Indeed the fractional linear map sends the domain to the first quadrant, the diameter to the positive real axis, and the arc to the positive imaginary axis. The normalized argument has the required boundary values. For $|z| \le \vartheta$,

$$\arg \frac{1+z}{1-z} = \arctan \frac{2 \operatorname{Im} z}{1-|z|^2} \le 2 \arctan \vartheta.$$

Apply the two-constants principle to the logarithm of each scalar functional of f_ℓ with functional norm at most one. It gives $\ell^{-1} \log \|f_\ell\| \le -a + (a+b)\omega$. Taking the supremum over these functionals is valid by the norm-dual formula. Approximation by smaller half disks handles absence of boundary continuity on $|z| = R$. Reflection of the domain handles the lower half; no reality condition on f_ℓ is required. The real diameter already satisfies the stronger estimate. □

For the source, ℓ is pattern size, $R \asymp s^{-2}$ and $a = c_0 R_0 - \log C$ after the real prefactor is incorporated. The complex b can depend on the fixed collar constant R_0. First choose R_0 to make $a > c$, then choose ϑ. There is no assertion that the resulting ϑ is bounded below uniformly as $R_0 \to \infty$. The same common field tube, regulator and fixed-entry Banach norm must be used on both the real interval and the complex disk. Comparing unrelated norms would invalidate the proof. This proposition makes a classical step explicit; it is not credited as a new maximum principle or as the main contribution.

Lemma 6.2 (Secants on an actual analytic disk)**.** *Let $f : D_R \to \mathcal{B}$ be holomorphic with* $\sup \|f\| \le M$*. For $0 < \varepsilon < R$,*

$$\left\| \frac{f(-\varepsilon) - f(0)}{-\varepsilon} - f'(0) \right\| \le \frac{M\varepsilon}{R(R-\varepsilon)}. \tag{30}$$

Proof. Cauchy's formula gives $\|a_n\| \le M/R^n$ for the Taylor coefficients, by first using radii smaller than R and then taking their supremum. The norm of the terms with $n \ge 2$ in the secant is at most $\sum_{n\ge 2} M\varepsilon^{n-1}/R^n$, the right side. □

This lemma exposes a relevant obstruction: at the large physical entry side $s \asymp \varepsilon^{-1/3}$, a bounded analytic function on a disk $R_s \asymp s^{-2}$ only gives a secant error of order $\varepsilon s^4 \asymp \varepsilon^{-1/3}$, not a vanishing bound. Smallness of ε/R_s alone is insufficient for a first derivative approximation. The source instead transfers to a canonical entry near the logarithmic scale J, with radius

$$\rho_J = cJ^{-2}e^{-2\gamma J}.$$

Its constants are chosen so $\gamma C_* < .01$ and $J = C_* \log s + O(1)$. Then

$$\varepsilon\rho_J^{-2} = O\Big(J^4 e^{(-3/C_* + 4\gamma)J}\Big) \longrightarrow 0.$$

After the separate physical readout and slaving comparisons, this supports the source's late-entry secant. It does not identify the different remainder Banach spaces, or manufacture (5).

7 From original Ram proofs to this report

The freshly read original stability text [5] proves the following general residual principle. If $F : X \to Y$ is continuously Fréchet differentiable on an admissible closed ball, $B : Y \to X$ is a bounded isomorphism, and

$$\eta = \|BF(c)\|, \quad q = \sup_{\|x-c\|\le r} \|I - BDF(x)\| < 1, \quad \eta + qr \le r,$$

then $x \mapsto x - BF(x)$ is a contraction of the ball into itself. The Banach fixed-point theorem gives a unique root there and $\|x - c\| \le \eta/(1-q)$. The isomorphism assumption is material: a merely rectangular left inverse can certify $BF(x) = 0$ without certifying $F(x) = 0$. Its perturbation theorem similarly presupposes existence of the perturbed solution in the controlled ball.

For the XY passage problem, an unqualified ball theorem on a long time interval can pay a large propagator norm. The present phase coordinate uses the specific hyperbolic invariant to replace that generic estimate by (10), and its discrete version pays (21). This is not a new fixed-point theorem. If numerical approximations rather than actual states are used, an appropriate existence/enclosure argument is still required; a residual computed only for a surrogate polynomial does not establish that the full RG activities follow it.

The freshly read original TR018 passages [6] require common connected analytic domains, fixed branches and locally uniform majorants for moment identities and their continuation. They explicitly separate continuation from convergence of the original integral and finite triangular inversion from infinite, uniformly bounded inversion. Those distinctions directly constrain Proposition 6.1 and Lemma 6.2: no finite interpolation table certifies a complex parameter disk unless the whole-disk majorant is supplied. Likewise, a finite nonsingular Jacobian or a tiny floating-point phase residual is neither a positivity certificate for P nor a uniform lower bound on K over all near-critical parameters.

8 Exit and mass interfaces that remain open inputs

The strongest justified interpretation is modular. Once actual states, true residual bounds and positive branches have been certified on a finite window, our theorems certify its clock. Extending this to (9) requires all of the following interfaces.

Entry and transverse response. The activity map must be the one representing the cosine model with its real boundary conditions and complex normalizers. The uniform regulator estimates, companion critical trajectory, canonical positive P and readout transfer remain source inputs. Bounds from finitely many temperatures cannot be promoted to uniformity as $b \uparrow b_c$.

The outgoing part. A compact-window certificate ends before the pole. For the exact solution with fixed κ, the time remaining after $X = -D < 0$ is

$$T - w_D = \frac{1}{\kappa} \arctan \frac{\kappa}{D}, \qquad 0 \le \frac{1}{D} - (T - w_D) \le \frac{\kappa^2}{3D^3}. \tag{31}$$

This follows from $X = \kappa \cot(\kappa w)$ and $u - u^3/3 \le \arctan u \le u$. It is an exact-flow formula only. For actual RG states the source uses its outgoing invariant cone and contracted activities, not a substitution of a finite residual into the singular endpoint. In its variables $D = d/\delta$ grows, while the unscaled coordinates must still stay inside the small tube.

The height and mass parts. Gaussian profiles with finite pins require probability normalization, uniform integrability and the ordered band/cable limits. Nonzero-total density smears in a pinned domain are valid and important; they cannot be replaced throughout by zero-mean tests. The lower mass comparison uses a nonvanishing separated pinned covariance and a large-loop bound; the upper comparison uses many annuli, arbitrary fixed polynomial decay and a finite-size contraction. We have not replaced either one by an ODE phase estimate.

Proposition 8.1 (Conditional error separation)**.** *Suppose for a family $b \uparrow b_c$ one has independently established numbers $j_e(b)$, T, and error bounds*

$$|\delta j_e - T| \le E_{\mathrm{flow}}(b), \qquad |\delta \log_L \xi(b) - \delta j_e| \le E_{\mathrm{mass}}(b).$$

Then

$$|\delta \log \xi(b) - T \log L| \le \log L \, \{E_{\mathrm{flow}}(b) + E_{\mathrm{mass}}(b)\}.$$

The clock theorems may contribute to E_{flow} only within their stated branch domains; they provide no E_{mass}.

Proof. Use $\log \xi = (\log L) \log_L \xi$ and the triangle inequality. □

This elementary separation prevents a finite-branch success from being reported as a measurement, proof or improved value of A_{XY}. No numerical value of that amplitude is asserted here.

9 Advantages, costs and next mathematical steps

The notable local gain is the corrected clock, not the classical cotangent solution by itself. It removes the linear x_j contribution from the leading discrete passage-time defect, admits an explicit constant, and extends to actual perturbed steps without an exponential in the number of steps. The

continuous and discrete versions share the same branch geometry, and the panel formulas expose exactly which finite arithmetic can be certified.

There are real costs. Near the separatrix the factors $1/k$, $1/(k\underline{k})$ amplify errors; positivity cannot be assumed from proximity in an unrelated norm. Conservative absolute budgets lose cancellation, as Example 3.3 shows. The invariant-product ledger can itself become poorly scaled if many $1 - y_j^2$ factors are small. Endpoint-only time brackets additionally need phase monotonicity. The fixed constants 7 and 2/5 were chosen for transparent proofs, not for optimal acceptance regions.

Useful future work is specific: obtain certified signed activity remainder projections in (19); combine adaptive panels with the full Banach activity enclosure; exploit exact time-change components before taking absolute values; and develop a compatible outgoing-cone certificate in the unscaled coordinates. A quantitative mass-comparison modulus, if proved for the true XY measure, could then be inserted in Proposition 8.1. None of these inputs is promised by the present calculations.

10 Evidence and contribution boundary

The full cached source consists of ten TeX files (4236 lines) and an 18-line README. All 4254 lines were freshly displayed in bounded, untruncated chunks and confirmed in the owned read ledger. This includes both figure source files and the bibliography; it does not mean that every cited external theorem or companion manuscript was audited. Pre-read hashes were captured before those displays, and post-read hashes match. Parent snapshot authentication is provenance only.

The fresh private-original scope is all 295 lines of the stability chapter and TR018 lines 1155–1244 and 1713–1840 (218 lines). The latter's unread complements are 1–1154, 1245–1712 and 1841–2244. No whole-corpus reading is claimed, and no private text or private title was submitted to a public search. The exact independent script checks polynomial identities, rational branch/clock inequalities, panel coefficients and subdivision, invariant-product propagation and the worked rational examples. It does not run a foreign checker, evaluate a complete RG map, simulate XY spins or certify the BKT amplitude. Parent checks are not counted as author checks. The author suite passed 2888 parameter cases with 39467 exact assertions using rational arithmetic and symbolic identities. It ran with the existing local Anaconda Python/SymPy installation; the bundled Python's missing SymPy caused an initial import failure, not a failed mathematical check. No package installation or shared-environment change was needed.

No mathematical correction to the cached source is established by this report. The source already has the balanced coordinates, controlled analytic entry, the cotangent passage and the separate mass argument. The new local results are Theorems 3.1, 4.2 and 4.3 with their explicit certificate interface. Elementary invariant integration, the Bernstein hull, Cauchy's formula, the maximum principle and the Banach contraction theorem retain their classical attribution. Applicability remains L3; contribution C3 reflects the demonstrated source-specific discrete refinement and its proof, not a claim of global theorem verification or historical priority over all numerical ODE literature.

References

[1] OpenAI mathematical manuscript collection, *Essential Singularity of the Correlation Length in the Planar XY Model*, October 5, 2026. Assigned cached source, pinned base `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Scientific sections: finite, entry, trajectory, profile and mass. Snapshot provenance and literal-read hashes are in the accompanying audit.

[2] Same public collection, *The critical correlation exponent of the planar XY model*, September 24, 2026, and *The critical logarithmic correction for the planar XY model*, October 5, 2026. Companion inputs cited by the assigned source; not freshly audited here.

[3] P. Lammers, *Bijecting the BKT transition*, arXiv:2301.06905v2, April 18, 2023. Primary Theorem 1 and normalization statements inspected. https://arxiv.org/html/2301.06905v2.

[4] T. Durand and P. Lammers, *The BKT transition and surface tension differentiability*, arXiv:2605.28473v1, May 27, 2026. Primary Theorem 1.4 inspected; not a complete independent proof audit. https://arxiv.org/html/2605.28473v1.

[5] M. Subbarao, original private Ram nonlinear integral-equation/PDE stability chapter, local file `TR1-lin-nonlin/stability.tex`. Fresh literal scope: lines 1–295. Confidential original proof material; not uploaded and not represented as a public publication.

[6] M. Subbarao, original private Ram Foundation TR018, `TR-rev-1-old-tr018.tex`. Fresh literal scope: lines 1155–1244 and 1713–1840. Analytic continuation, uniform majorants and finite-inverse limitations; confidential original.